



BSI Standards Publication

Endoscopes — Medical endoscopes and endotherapy devices

Part 4: Determination of maximum width of insertion portion

National foreword

This British Standard is the UK implementation of ISO 8600-4:2023. It supersedes BS ISO 8600-4:2014, which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee CPW/172, Optics and Photonics.

A list of organizations represented on this committee can be obtained on request to its committee manager.

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Date	Text affected
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**Endoscopes — Medical endoscopes
and endotherapy devices —**

**Part 4:
Determination of maximum width of
insertion portion**

*Endoscopes — Endoscopes médicaux et dispositifs d'endothérapie —
Partie 4: Détermination de la largeur maximale de la partie insérée*





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Foreword

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The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

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This document was prepared by Technical Committee ISO/TC 172, *Optics and photonics*, Subcommittee SC 5, *Microscopes and endoscopes*.

This third edition cancels and replaces the second edition (ISO 8600-4:2014), which has been technically revised.

The main change is as follows:

- [5.2](#), Capsule endoscope, was added.

A list of all parts in the ISO 8600 series can be found on the ISO website.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

Endoscopes — Medical endoscopes and endotherapy devices —

Part 4: Determination of maximum width of insertion portion

1 Scope

This document specifies a method of measurement of the maximum insertion portion width of medical endoscopes and certain endoscopic accessories.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 8600-1, *Endoscopes — Medical endoscopes and endotherapy devices — Part 1: General requirements*

ISO 8600-6, *Endoscopes — Medical endoscopes and endotherapy devices — Part 6: Vocabulary*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 8600-1, ISO 8600-6 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <https://www.electropedia.org/>

4 Test conditions

4.1 Test environments

The test environment conditions shall be as follows:

- temperature from 15 °C to 35 °C.

NOTE Relative humidity and atmospheric pressure is not defined because measurement devices made of metal or plastic are not affected by them.

4.2 Accuracy of measuring instruments

Measuring instruments with a minimum accuracy of 0,05 mm shall be used (e.g. by a vernier caliper).

For measurement of the peripheral length, in French size, measuring instruments with a minimum accuracy of 0,5 mm shall be used (e.g. by a tape measure or a similar tool).

5 Method of measurement

5.1 Flexible and rigid endoscope

5.1.1 General

For measurement of the millimetre indication, the maximum diameter of a circumscribed circle perpendicular to the nominal axis of the insertion portion shall be measured. See [Figure 1 a\)](#) and [Figure 1 b\)](#). This maximum diameter is defined as the largest diameter measured in all sections perpendicular to the nominal axis along the length of the insertion portion.

For measurement of the French size indication, the maximum peripheral length of a section perpendicular to the nominal axis of the insertion portion shall be measured. The maximum peripheral length is defined as the largest peripheral length measured in all sections perpendicular to the nominal axis along the length of the insertion portion.

Flexible endoscopes shall be measured with the insertion portion straight.

5.1.2 Diameter indication, d

To obtain the diameter of an endoscope:

- measure the maximum diameter of a circumscribed circle, in millimetres;
- if an endoscope utilizes a detachable hood, measure the maximum diameter of the endoscope both with and without the detachable hood. See [Figure 1 a\)](#) and [Figure 1 b\)](#);
- the unit of measurement shall be millimetres.

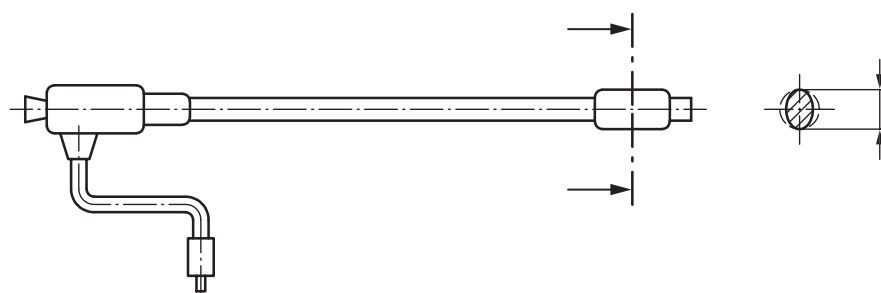
5.1.3 French size indication, Fr

To obtain the French size indication:

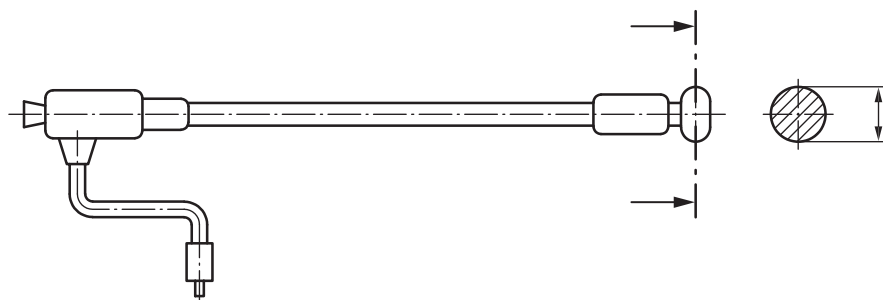
- if the section of the insertion portion is circular, the French size is calculated by multiplying the measured diameter by three;
- if the section of the insertion portion is noncircular (see [Figure 2](#)) measure the minimum length, u of the circumscribed curve and calculate the French size, Fr , by [Formula \(1\)](#):

$$Fr = \frac{3u}{\pi} \quad (1)$$

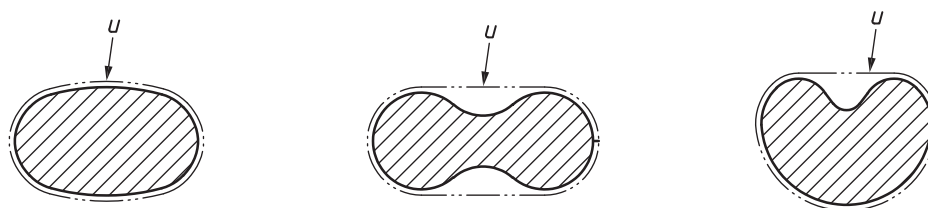
where u is the minimum length of the circumscribed curve, in millimetres.



a) Without detachable hood



b) With hood

Figure 1 — Examples of measurement of maximum diameter of a flexible endoscope**Key**

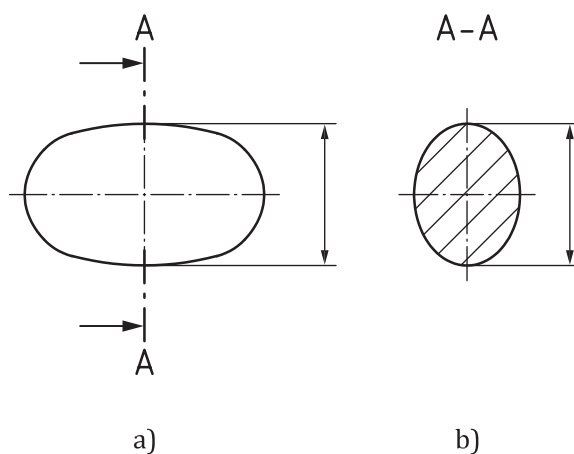
u is the minimum length of the circumscribed curve, in millimetres

Figure 2 — Examples of non-circular insertion portion sections**5.2 Capsule endoscope**

For measurement of the millimetre indication, the maximum diameter of a circumscribed circle perpendicular to the nominal axis of the longer direction of the capsule endoscope shall be measured. See [Figure 3](#). This maximum diameter is defined as the largest diameter measured in all sections perpendicular to the nominal axis along the longer direction of the capsule endoscope.

To obtain the millimetre indication:

- measure the maximum diameter of a circumscribed circle;
- the unit of measurement shall be the millimetres.

**Figure 3 — Example of measurement of the maximum diameter of longer direction of a capsule endoscope**

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